

An IoT Enabled Air Pollution Meter with Digital Dashboard on Smartphone

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Abstract— Air Pollution which is environmental as well as a social issue faced by the individuals, since it has a variety of negative consequences for human health, climate change and ecosystems. Today, one of the Europe's most important environmental health hazards is Air Pollution. The air quality is the prominent aspect which directly increases the prevalence of illness and reduces the quality of life in cities and metropolitan cities. The ability to make respective solution with in the particular time depends on the air quality data collection and interpretation, necessitating the implementation of monitoring system that monitors the air quality periodically. A full level assessment of key contaminants and their sources can be accomplished using multi-parameter air quality monitoring devices. These monitoring devices are used in various smart cities efforts to monitor air quality as well as minimizing the concentrations of the major polluting substance in various urban and sub-urban cities. The design for detecting air pollution and promoting public awareness is included in this proposed system. The purpose of this research is to leverage the Internet of Things (IoT) in conjunction with the cloud to provide services much more in real-time as well with improved speed. The proposed device will be used in a specific place where there is a lot of pollution. The level of each hazardous pollutant is tested at regular intervals. The amount of each detected pollutant as well as the air quality index at that specified location will be displayed on the smartphone, which will be utilized to calculate the Air Quality Index which is abbreviated as AQI for the perceived pollutants and enhance public awareness. As a result, by examining the gas concentrations in both numerical and graphical form, the public can have a better understanding of the air quality in that place. Furthermore, in the future, the public will be able to register in an app that will send out weekly or monthly air quality reports via a message that will reach the user as an easier-to-access alert.

Keywords— Air quality, Pollution, Arduino, Air Quality Index,

I. INTRODUCTION

Air Pollution is the most serious environmental issue, with several negative consequences for human health, water bodies and climate. Vehicles are the primary source of air pollution in all large cities, with industries as the secondary source. The widespread usage of automobiles has resulted in a significant increase in the amount of pollutants in the environment. This is the source of environmental pollution that has a negative impact on human health.

Other respiratory issues such as asthma attacks and skin rashes, have also arisen as a result. The Central Pollution

Control Board has established a standard for these levels, but the general population is hesitant to adhere to it. The pollutants that pollute the air is invisible, which has contributed to people's indifference. As a result, today's primary requirement is public recognition. As a result, the suggested system resolves this key problem. The air pollution monitoring system is deployed in a specific location where there are indications of acute air pollution in order to detect the constituent gases of air that may cause harm to human health and other living species.

This system predicts the levels of numerous dangerous gases such as CO, NH₃, particulate matter and smoke using Arduino and multiple gas sensors. Carbon monoxide is given first priority because it is a greenhouse gas and a major pollutant that contributes to global warming, previous products did not have the benefit of keeping pollution rates current. As a result, an IOT air pollution meter was employed to solve the problem. The measured air quality level is also displayed on the smartphone screen, allowing users to stay up to date on current air quality. Users can see the air quality level in both numerical and graphical forms. In addition, the application calculates and displays the Air Quality Index (AQI) for the present pollution level, as well as health impacts. As a result, this tool enables users to take effective precautions ahead of time to protect themselves from negative consequences.

II. LITERATURE REVIEW

Development of a Bluetooth-enabled health monitoring system for infants that uses wearable technologies. The creation of a wearable monitoring device that senses and displays an electrocardiogram (ECG) and core body temperature is detailed in this paper. A conductive cloth is used to obtain an ECG, and a thermistor-type temperature sensor is used to measure temperature. The invention accepted is a practical application of the Internet of Things concept [1].

Lora wireless network system which is based on Internet of Things (IoT) is designed and developed for monitoring of human signal. My-signals development shield is used in conjunction with a LoRa wireless network system to create an IOT-based health monitoring system to create an IOT-based health monitoring system, My-Signals and LoRa have employed a body temperature sensor, a pulse rate sensor, as well as a sensor to measure the saturation of oxygen and an electrocardiogram (ECG) sensor [2]. this paper discusses on the construction and development of air quality monitoring

system which is based on IOT. The current research presents a field-deployable air quality monitoring system that uses the Internet of Things (IOT) to analyse real-time data on levels of carbon monoxide, smoke and particulate matter. The device can detect air pollution in a given location and generate analytical data [3].

In the sphere of ecological insurance, the key role is played by the Air Pollution Monitoring as well as Forecasting System which is based on IOT. It also describes the system for continuously checking and determining air pollution [4].

Employing the Wireless Sensor Network in order to monitor the air pollution provides two whole number straight programming strategies that are based on real poisonous scattering displays. In this case, we consisted two design models in the aspect of their implementation season and the second low-based definition is far superior [5].

Information about the system is provided through a wireless sensor network in order to continuously monitor the air pollution rate, which consists of a large number of appropriately configured observation stations that communicate with a back-end server via machine-to-machine communication [6].

III. POLLUTION METER SET UP

A. Pollution Meter Specifications

A product necessities detail (SRS) is a representation of a product framework that will be built it is defined after business requirements are determined, also known as partner requirements specification.

B. Hardware Requirements

1. Arduino Uno

The ATmega 328 microprocessor is employed by the Arduino Uno microcontroller board (datasheet), 6 analog inputs along with a 16 MHz resonator, a power connector a USB port, an ICS header, and a reset button are present on the respective board along with the availability of 14 digital pins which is used for input and output process, then the other 6 pins is utilized for connecting with the PWM outputs. In order to get implemented, first connect it to the system with the help of a USB cable, else it can be done with the help of battery or an AC to DC adapter in order to power it.

The FTDI USB to serial driver chip available on earlier boards is not employed in this Uno, which makes it distinct from the other boards. Instead, it makes use of a USB to serial converter which is based on the ATmega16U2. The revision 2 Uno board's 8U2 HWB line is then pushed to the ground with the help of a resistor, making it simple to enter DFU mode.

2. Dust Sensor

Dust Sensor is a straightforward air monitoring gadget with a sharp G2Y1010AU0F on-board. It can detect fine particles with a diameter of more than 0.8m, such as cigarette smoke. The sensors analogue voltage output is proportional to dust density. The module includes an inbuilt voltage boost circuit that allows it to work with a wide range of power supplies.

3. DHT11 Sensor

The DHT11 is one of a vital computerised temperature as well as a dampness sensor that saves time and effort. It measures the surrounding air using a capacitive moisture sensor as well as a thermistor, then the outputs an advanced signal available in the data pin. It is really handy to employ, but it requires careful planning in order to obtain information. The main genuine disadvantage is that it can acquire only the new data once for on every standard span, hence sensor readings could be outdated for maximum to 2 seconds while using this required library.

4. Gas Sensor

A frequently utilized as a component of a safety system, a gas detector is one of the component that is deployed to detect the quantity of gases which is present in specific region. This kind of technology can connect with a control system that can automatically stop an operation while also detecting pollution or gas leaks. A gas detector would notify operators in the vicinity of the leak, giving them the chance to leave. This kind of technology is required since many gases could be harmful towards organic life, such as people or animals. Gas detectors can identify smouldering, flammable, toxic and oxygen-deficient gases as well as these conditions.

This kind of machinery is frequently used in industry, and it can be seen in locations like oil rigs to keep an eye on production process and emerging technologies like photovoltaic. To put out flames, they are used.

5. GSM

GSM which is "Global system for mobile communication" refers to a mobile communication modem. Data can be processed by the digital system at rates which ranges around 120 Mbps.

In a Macro, micro, Pico and umbrella cells are some of the types of cells employed in the GSM network. According to the domain of implementation, every cell is unique. The access area of each type carries depending on the implementation environment.

6. ES8266 Module

Any microcontroller can connect to your Wi-Fi network with the ESP8266 Wi-Fi Module which is a self-contained SOC. Both hosting an application and offloading the Wi-Fi networking functions to an additional application processor are options for the ESP8266. Each ESP8266 module is already pre-programmed with the AT command, therefore all you need to do is put it into your Arduino to start using it as a Wi-Fi shield (out of the box). An affordable board with a sizable and expanding community is the ES8266 module.

C. Software Requirements

1. Arduino IDE

The cross-platform Arduino integrated improvement condition application was built in Java. With the aid of external centres and other commercial improvement sheets, it may be used by both beginners and experts to build and transfer projects to Arduino good sheets.

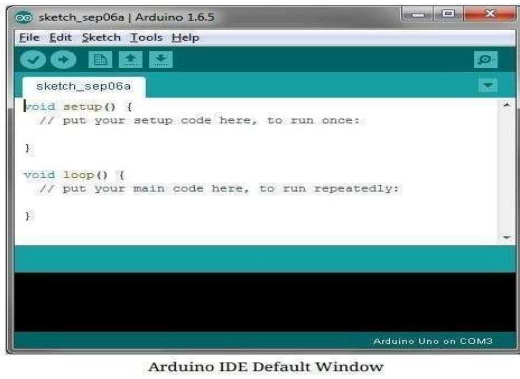


Figure 1 Arduino IDE Default Window

2. Pre-requisites for programming

The Arduino – The C++ and C programming languages were used to create the multi-platform Arduino Integrated Development environment, which runs on OS such as Windows, Linux as well as macOS.

The IDE's source code is made available under the General Public License with a rendition. It is utilized to create and move projects to Arduino flawless sheets, as well as other vendor improvement sheets, employing pariah focused.

3. Embedded C

Embedded C is one of the programming language that is employed for designing electronic devices. To perform the prescribed tasks, the CU needs embedded C code. We employ various technological tools in our day to day life, such as a cell phone, camera as well as washing machine. The gadgets are all controlled by a microcontroller which is coded with embedded C code.

C code is preferred over other languages in embedded system programming. Because of the following factors (i) Easy to understand (ii) High reliability (iii) Portability (iv) Scalability

IV. PROPOSED SYSTEM

A. Block Representation

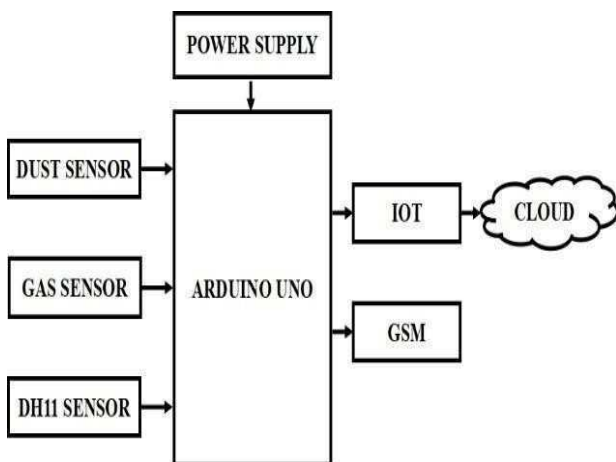


Figure 2 Block Representation of Proposed System

B. Working of the Proposed System

IOT Application for contamination executives, business based expressly, smooth out and incorporate natural procedures such as emission of air, energy as well as water executives. Such applications allow clients and business visibility into the dangers of events like chemical spills, oil slicks and hazardous chemicals improper removal, while maintain consistency with natural gauges and rules.

Advancement re-appropriate assistance for such applications, which can include areas such as

1. Across associations, there is a common stage to monitor and oversee activity conditions.
2. Real-time natural procedures see, review and discover occurrences and discoveries in accordance with ISO 14001 and other industry explicit natural principles.
3. Risk aspects associated with limit breakers.
4. Trigger notices for edge penetrated the application for monitoring the quality of air had the following features
 - a. Air quality indices for particular area based on ongoing calculations, air quality day by day gauges.
 - b. Timing open air exercises for various age groups
 - c. Air quality drops associated with health hazard.
 - d. Related reports for the quality of air estimates based on areas
 - e. Air quality maps age.
5. This project is an IOT enabled air tainting metre that monitors air quality on your phone using the server and an Arduino board. The server stage of the Internet of Things is often used to operate Arduino boards. On your phone, the server delivers a computerized dashboard that displays continuous air quality data for your immediate area.

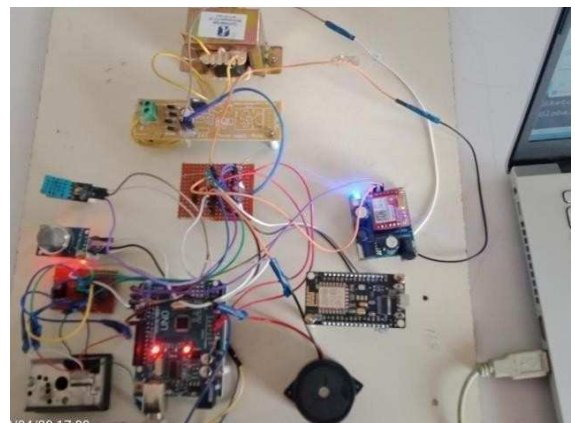


Figure 3 Prototype of Proposed System

6. The IOT server isn't made to function with a certain board or shield. Whether you use Wi-Fi, Ethernet or an ES8266 chip to connect the Arduino or Raspberry Pi to the Internet, it will prepare you for the web and the Internet of Things. The analogue and digital pins

of the Arduino are used to connect the sensors. The microcontroller has an ADC built in.

7. An ADC is a data processing mechanism that converts analogue signals into digital codes, allowing originality and digital circuitry to coexist. The values of analogue signals change all the time and are collected by an array of sensors that track a variety of variables. Many digital devices communicate with their surroundings through analogue signals. Through the transmitter and receiver, the data from the analogue to digital converter is sent to the Wi-Fi chip (Tx and Rx).
8. After then, the data is stored on a private cloud platform. The open source mobile application can keep track of the data stored in the cloud. Warning and frequent alerts are given when the values for Hardware set up exceed the threshold value.
9. The air pollution monitoring plan structure has three main stages
 - a. Stage 1 – Using sensors, identify the clustering of air pollutants in the intrigue zone.
 - b. Stage 2 - Provides user-friendly environment and compact interface so that the client can utilize to determine the level of contamination in his or her unique territory.
 - c. Stage 3 – Employing the logical module, predict air quality. The framework for monitoring of air pollution that has been presented.

C. Functional Requirements

The DHT11 sensor will collect temperature variation and humidity of various devices, and the gas sensor will capture information about gases floating in the air and notify the end user about their devices available via text notification.

D. Air Quality Monitoring

Input – The DHT11, gas sensor and M 2.5 sensor all provide readings. The analogue readings from the DHT11 sensor are taken using an Arduino Uno.

Processing – The Arduino Uno collects the signals from the DHT11, gas sensor and M 2.5 sensor and compares them to the device's threshold value.

Output – Effectively monitor temperature deviations, gases floating in air, and liquid particles in the air as well as inform end users about their devices via smartphone.

V. RESULTS AND TABULATIONS

A. Data Analysis

1. **Input** – The device's air pollution and the device threshold value.
2. **Processing** – Analyzing the data collected by the sensor and the device's threshold value.
3. **Output** – Air quality near the gadget and smartphone display.

B. Performance Expectations

1. A reliable internet connection
2. A cache-managing browser.
3. A functional DHT11, as well as a gas sensor and a PM 2.5 sensor.
4. An Arduino Uno Microcontroller board.

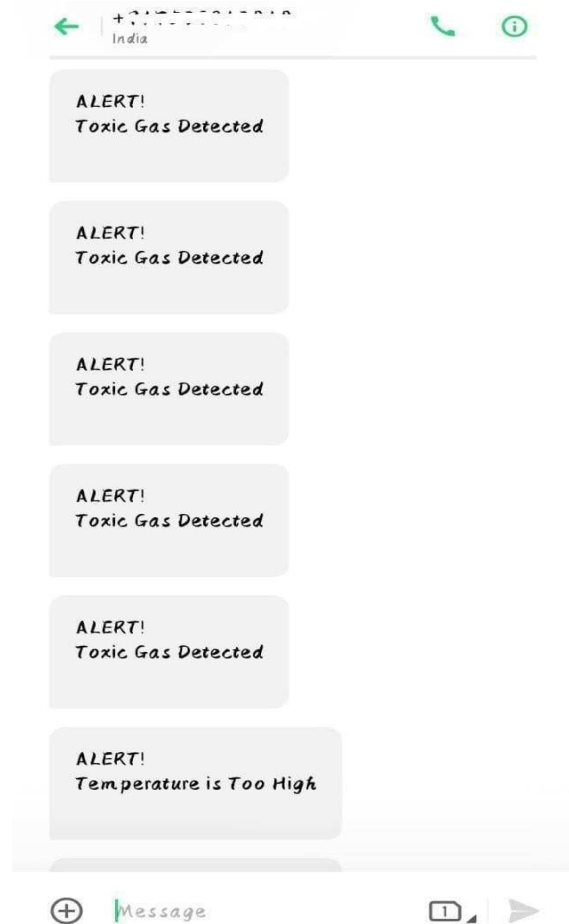


Figure 4 Alert Message

C. Proposed Model Mathematical Analysis

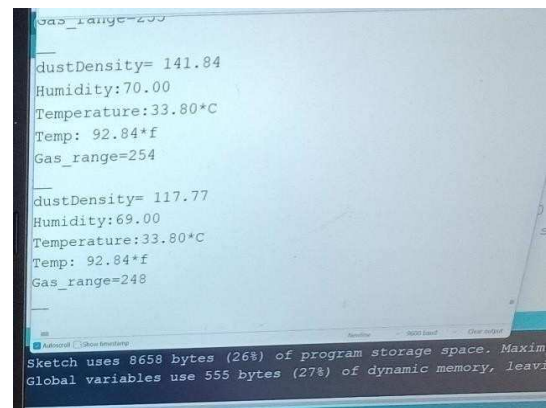


Figure 5 Output in the Application

Pollutant concentrations in the air is quantified in terms of parts per million as well as in percentage. The sample conversion

$$1.145 \text{ mg/m}^3 = 1 \text{ ppm } 1 \text{ mg/m}^3 = 0.873 \text{ ppm}$$

$$1 \text{ percent} = 1/100 \text{ parts per million} = 1/1000000 = 0.0001 \text{ percent Equals 1 part per million}$$

VI. CONCLUSION

Within the estimation of a person's typical life expectancy, the researches created an electronic monitoring system based on the indoor oxygen concentration for the study. The temperature, relative humidity, gas concentrations, and oxygen levels in the area were all being monitored by this equipment. The Dust sensor, DHT22 and MQ4 sensors connected to the Arduino are used to read these activities. A Wi-Fi module is used by desktop, web and mobile programmes to obtain the readings via a serial connection. With the use of this technology, it is hoped that prisoners would be able to observe how long they have lived and discover that it has been longer than they had anticipated. They will live longer as a result. This particular device has provided us with the system that enables us to stay connected to it while also altering people who can perform emergency interventions in situations like a decrease in life expectancy. This system enables registered data to be listed with details and being monitored on a regular basis, resulting in accurate and timely intervention. By using local resources to build this prototype, we can enhance our country's economy by reducing our reliance on outside resources.

With the correct funding, this prototype technology may be easily produced in our country. The ability to make equivalent goods will be improved as a result of this. This technology that has been developed also carries the responsibility of directly saving lives. These scientific domains and other multidisciplinary research can make use of human psychology, which is informed about the rate at which oxygen remains in the inner environment and the

person's behavioural pattern who visualizes the instant amount of oxygen among other things.

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